

VPC Flow Logs

- VPC flow logs help you capture and monitor network traffic (IP traffic) going to and from network interfaces in your VPC.
- It is used to check what traffic is coming in and going out of your VPC, subnet, or EC2 instance.
- The collected logs can be sent to Amazon CloudWatch.

- Logs or Amazon S3 for viewing and analysis.
- once created, you can see and download flow logs from your selected destination.
 - No agent is required. it works without installing anything on EC2 (agentless).
 - You can enable flow logs at the vpc level, subnet level, or individual network interface level.
 - it basically records the traffic flow inside your vpc.

Aggregation time interval

- The aggregation interval is a time period during which traffic is grouped together into one flow log record.
- By default, AWS groups traffic for 10 minutes before creating a log entry.
- You can change this to 1 minute, but this will create more data (more log records).
- After the aggregation period ends, it takes some extra time to process and deliver the logs.
 - for cloudwatch logs: around 5 minutes
 - for S3: around 10 minutes.

VPC Flowlog fields

<u>Fields name</u>	<u>meaning</u>
Account ID	The AWS account that owns the resource (EC2, VPC etc)
ENI ID	The Elastic network interface (ENI) where traffic is being captured.
Source IP	The IP address sending the traffic (origin of the request)
Destination IP	The IP address receiving the traffic (target machine)

Source port	The port number on the source machine used for communication
Destination port	The port number on the destination machine (eg. 80 = HTTP, 443 = HTTPS)
Protocol	Protocol number (eg. 6 = TCP, 17 = UDP, 1 = ICMP)
Packets	Number of data packets sent in this connection
Bytes	Total amount of data transferred in bytes
Start time	Start of the traffic flow (Unix time format)
End time	End of the traffic flow (Unix time format)
Action	Whether the traffic was ACCEPTED or REJECTED by security rules.